

Practice Midterm, Summer 2024

This test has **5** questions worth a total of **100** points.

Read the submission instructions below carefully before starting the exam.

1. First, read through the exam PDF carefully, and answer the questions you feel most comfortable about first.
2. The exam consists of five short-answer questions. You **must** submit your answers as a **typed PDF**, with **one question response per page**. That means that you should submit a five page PDF, with your answer to Question 1 on page 1, your answer to Question 2 on page 2, etc.
3. Just to make my life easier, at the top of each page you should write the question that you are answering on that page. For example, your first page should have **Question 1** written at the top.
4. The response to each of the five questions can be given as (at most) a short paragraph. Excessively long answers will be **penalized**.
5. If you fail to follow these submission guidelines (in addition to just making your exam harder to grade) I will **deduct** a few points from your final midterm score.
6. You **must** submit your PDF to this assignment on Canvas by 8:25PM (the end of class) on the day of the exam: any late exams will not be graded.
7. You **are** allowed to resubmit to the Canvas assignment as many times as you like if you find a mistake (**as long** as your final submission is before the end of the exam!)

1. Should This Work? [20 Points]

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
def mystery(funky):  
    for i in range(1000):  
        print("funky")  
        print("meow")  
  
    for i in range(100):  
        funky += 1  
        meow += 3  
  
oof = 100  
meow = 50  
mystery(oof)
```

2. What Do I Need to Change? [20 pts]

The code below is a broken version of the solution to Lab 4, Question 2. Why is this solution broken, and what would we need to change in order to fix things? There may be more than one issue with the program: you should identify and fix all of them.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
# TODO: Get the number of books purchased from the user.
number = int(input("How many books did you buy: "))

# TODO: Determine how many points were earned.
if number >= 0:
    points = 0
elif number >= 2:
    points = 5
elif number >= 4:
    points = 15
elif number >= 6:
    points = 30
elif number >= 8:
    points = 60

print('This earns you this many points:')
# TODO: Print the number of points earned (just the number).
print(points)
```

3. What Do I Need to Change? [20 pts]

The program below is intended to read in a number from the user, add 100 to it, and print the result. Why will this program not work? What needs to be changed about this program to get it to work? There may be more than one issue with the program: you should identify and fix all of them.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
number_from_user = input("Enter a number please: ")
result_number = number_from_user + 100
print(Result_number)
```

4. Functionally and Fundamentally Quite Fun [20 Points]

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
def fun_function(first, second, third):  
    print(first)  
    print(second)  
    print(third)  
  
    print(x)  
    print(y)  
    print(z)  
  
x = 100  
y = 1000  
z = x + y  
fun_function(x, y)
```

5. Scrambled (Like an Egg)

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
j = 0

for i in range(10000):
    print(i)
    j += 1
    j *= 2

print(x)
x = j + 7
```

You have reached the end.

There is no more.

Go outside after this and take a study break. You deserve it :)

